

Matthieu Vinet

Robotics Engineering Student — Polytech Sorbonne



Seeking a 6-month final-year internship · Robotics · Paris · 2027

Robotics engineering student at Polytech Sorbonne, comfortable across the whole stack — from software (ROS2, vision) to electronics and mechanics. I enjoy designing and prototyping complete systems, from concept to working prototype.

Paris, France

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EDUCATION

Engineering Degree (MEng), Robotics · Polytech Sorbonne

2024 – 2027

Paris, France

Engineering Preparatory Cycle (PeiP) · Polytech Tours

2022 – 2024

Tours, France

Scientific Baccalaureate (high-school diploma) · Lycée Isaac de l'Étoile

2019 – 2022

Poitiers, France — majors in Mathematics & Engineering Sciences, advanced Maths option

EXPERIENCE

Robotics Engineering Intern · Botshare.ai

Apr. 2026 – Jul. 2026

Berlin, Germany — on-site (Mechatronics Lab, BHT University)

- Designed and developed the precision docking system (wireless charging) of an autonomous mobile robot — camera + a bundle of 3 AprilTags, visual servoing and a 6-phase docking sequence (open-source OpenAMRobot project).
- Developed and documented the `openamrobot_docking` ROS2 package (Gazebo simulation, Nav2 integration), released as open-source.
- Wrote a technical article presenting the approach and design choices on the Botshare.ai blog.

Member then Treasurer · Robotech Association (robotics — Polytech Sorbonne)

Since Sep. 2024

- Member since 2024 and treasurer for one year (April 2025 – April 2026, handover completed); active in the club's revival and in several team projects (mobile robot, arcade cabinet, RC plane, DIY motor).

Wwoofing · Traditional farm (Norway), summer 2025 — manual work, cultural immersion and daily English practice.

Side jobs & volunteering · Crew member — McDonald's, Poitiers (summer 2022) ; volunteer — SPA animal shelter, Poitiers (summer 2023).

PROJECTS

Robotics (academic project): mobile robot with a control HMI and an arm / pantograph prototype (CAD SolidWorks + electronics), ROS2 navigation ; real-time object tracking (OpenCV) ; image processing.

Electronics & fabrication: DIY brushless motor, arcade cabinet, RC plane — prototyping at the FabLab (3D printing, laser cutting).

Programming & web: chess game in C ; poker AI using the Monte-Carlo method (Python) ; personal website matthieu-vinet.fr (HTML / CSS / JavaScript).

→ All my projects in detail at matthieu-vinet.fr

TECHNICAL SKILLS

Robotics: ROS2, Nav2, Gazebo, OpenCV, visual servoing, AprilTag ; kinematics, frame transformations, Jacobian matrix

Programming & web: Python, C, C++, JavaScript, HTML / CSS ; Git, Docker, Linux

AI: AI tools & LLMs (ChatGPT, Claude, GitHub Copilot) on a daily basis ; basics of reinforcement learning (Q-learning)

Electronics & embedded: breadboard prototyping, soldering, component selection, power electronics (DC-DC), sensors (LiDAR, IMU), DC & brushless motors (BLDC) ; Arduino, ESP32, Raspberry Pi boards

CAD & prototyping: SolidWorks, 3D printing, laser cutting (FabLab), Matlab

SOFT SKILLS

Curious · Rigorous · Determined · Self-driven · Adaptable · Fast learner · Team player

LANGUAGES

French — native · English — professional

INTERESTS

Video games · Programming · Prototyping & robotics